# DSA 8401 — Applied Machine Learning

## Capstone Project

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|---|---|
| Weight: | 60% of final grade (project, participation) |
| Mode: | Each individual student submits their own end-to-end system |
| Proposal due: | Week 10  28 September 2026 |
| Presentations: | Week 15  02 November 2026 |
| Lecturers: | Prof. Javier Serrano & Emmanuel Bundi |

# 1. Purpose

The capstone is the course's culminating deliverable. You choose any machine learning problem that interests you; the techniques you apply should be the ones that genuinely fit your problem.

Because the capstone carries 60% of the final grade, it is strictly an **individual** submission. Each student selects their own problems and dataset, designs their own architecture, and defends their own decisions.

# 2. Problem Selection

**Each student chooses their own machine learning problem and dataset.** You decide which ML techniques to apply use what fits your problem, not what fills a checklist. The only constraints are:

**Real-world messiness.** The data must be raw or semi-raw not a pre-cleaned Kaggle benchmark. If you use a public dataset, you must introduce a realistic complication (e.g., temporal splits, label noise, missingness, class imbalance) and justify it.

**Sufficient depth.** The problem must be complex enough to justify building and deploying an end-to-end system.

## 2.1 Suggested Problems

The following problems are anchored in the African tech and fintech ecosystem. You may choose one of these or propose an entirely different problem domain.

| Track | Example problem |
|---|---|
| Fintech / under-banking | Alternative-data credit scoring for thin-file applicants using mobile-money transaction logs |
| Trade-corridor underwriting | Risk modelling for cross-border SME lending along East-African trade corridors |
| Health informatics | Treatment-policy modelling, disease image classification, or clinical pathway analysis |
| Fraud & AML | Streaming anomaly detection on payment rails with concept-drift monitoring |
| NLP for African languages | Sentiment analysis, hate-speech detection, or MT on code-switched corpora |
| Agriculture & climate | Crop-yield forecasting, satellite image segmentation, or pest/disease early-warning |
| Other | Any problem approved at the proposal stage |

# 3. Technical Requirements

The three requirements below apply to every capstone regardless of which ML techniques you choose. Beyond these, you are free to use any combination of methods from the course or beyond it that fits your problem.

## 3.1 Minimum Requirements (must have)

Every capstone must include these three elements:

| # | Requirement | What this means |
|---|---|---|
| 1 | Reproducible ML pipeline | A scripted, end-to-end pipeline: data ingestion → preprocessing → model training → prediction. No notebook-only submissions. The pipeline must be runnable from scratch by a reviewer. |
| 2 | Model service & deployment | Your trained model wrapped in an API (FastAPI or equivalent), containerized with Docker, and served so a client can send data and get predictions back. |

## 3.2 Your Modelling Choices (student's call)

Which ML techniques you apply are entirely up to you. Pick the methods that genuinely fit your problem. The course covers a wide range and of the following are fair games, alone or in combination:

| Syllabus Area | Example techniques |
|---|---|
| Classical supervised learning | Logistic regression, SVM, random forests, XGBoost/LightGBM, cost-sensitive tuning |
| Unsupervised learning | K-Means, DBSCAN, PCA, t-SNE, anomaly detection |
| Deep learning | MLPs, CNNs, RNNs/LSTMs, Transformers, transfer learning |
| Generative AI | LLM fine-tuning (LoRA), RAG pipelines, agentic loops |
| Feature engineering | Temporal features, encoding strategies, leakage audits, SHAP explanations |
| Monitoring & fairness | Data/concept drift detection (Evidently AI), subgroup fairness metrics |
| MLOps extras | CI/CD pipelines (GitHub Actions), experiment tracking (MLflow), model registries |

In your report and presentation, you must justify why you chose the techniques you did and why you did not use others that might seem relevant. Thoughtful exclusion is as important as inclusion.

# 4. Deliverables Checklist

Each student submits a single Git repository containing all the following:

- ☐ README.md project overview, setup instructions, architecture diagram, how to reproduce
- ☐ AI_USE.md disclosure of any AI assistance, per course policy
- ☐ Data pipeline code (scripted, reproducible)
- ☐ Model training code and serialized artifacts for your chosen approach
- ☐ Docker file + docker-compose.yml (or equivalent)
- ☐ API service code (FastAPI or equivalent)
- ☐ Final report or slide deck: problem framing, technique justification, results, limitations

# 5. Assessment Rubric

60% capstone weight is distributed across the following components:

| Component | Weight | What we are looking for |
|---|---|---|
| Data pipeline | 10% | Clean, scripted, reproducible pipeline. Data preprocessing is appropriate and documented. |
| Modelling & evaluation | 20% | Chosen technique(s) are appropriate for the problem; evaluation is rigorous and honest; results are interpreted, not just reported. |
| Technique justification | 10% | Clear written reasoning for why you chose these methods and why alternatives were ruled out. |
| Deployment (API + Docker) | 10% | API works, container builds, deployment is reproducible from the README. |
| Final report & presentation | 10% | Clear communication of the problem, architecture, results, limitations, and real-world relevance. |
| **Total** | **60%*** | |

# 6. Proposal Format (Week 10)

Each student submits a **one-page proposal** (maximum 500 words) covering:

1. **Problem statement —** what you are predicting, detecting, or generating, and why it matters.
2. **Data source —** where the data comes from, approximate size, known quality issues.
3. **Planned approach —** which ML techniques you intend to use and why they fit this problem. Include a sketch of how you will serve the model.

The teaching team will approve, request changes, or suggest alternatives within one week.